your applications with React, Next.js, and the Vercel AI SDK. Toward the end, you'll see how to work with the Model Context Protocol to access external tools and data sources in a secure and standardized way.

I assume that readers have a basic understanding of JavaScript and familiarity with building simple React applications; everything else will be introduced as needed. Our focus is on the concepts and techniques essential for designing, developing, and deploying AI-powered web experiences.

Throughout the book, you'll work through focused, hands-on projects that gradually build your skills and confidence. In the final chapters, you'll apply what you've learned to create two real-world applications: an interview assistant that records voice input and provides AI-generated feedback and a more advanced corporate knowledge management system powered by retrieval-augmented generation (RAG). These projects reinforce key ideas while giving you strong portfolio pieces you can adapt for future work.

The goal of this book is simple: to empower you to build real, working generative AI applications. We'll start with a quick look at how generative AI works, along with its benefits, limitations, and uses.

1.1 What generative AI can do for web applications

Generative AI opens a wide array of opportunities across various domains.

1.1.1 Generative AI capabilities

AI can enhance web apps in many exciting and useful ways:

- *Text generation*—Producing human-like text for content creation, storytelling, and conversational agents. This is perhaps the most prevalent feature, as it all begins with crafting prompts using natural language and conducting follow-up conversations using tools like ChatGPT, Perplexity AI, Anthropic Claude, or Google Gemini.
- *Image generation*—Creating realistic images for artistic purposes, design, and creative projects. DALL-E and Midjourney are well-known image generators.
- *Multimedia generation*—Producing videos, music, and other multimedia content. OpenAI Sora 2 is popular for video generation.
- *Code generation*—Assisting developers in writing code, automating repetitive tasks, and enhancing productivity. GitHub's Copilot and Claude Code are two code generators in wide use.
- *Learning exploration*—Encouraging exploration and experimentation in creative fields, such as art, music, and literature.
- *Problem solving*—Tackling complex problems in various domains, including science, engineering, and medicine, through innovative solutions.
- *Content enhancement*—Improving existing content through editing, enhancement, and refinement. For example, you might generate images in cartoon or art styles like sketch, anime, or filters from a photo.